

Verification and Validation of openInjMoldSim, an Open-Source Solver

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Submitted by

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DEPARTMENT OF MECHANICAL ENGINEERING

INDIAN INSTITUTE OF TECHNOLOGY KHARAGPUR

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DECLARATION

The project thesis titled “ Verification and Validation of openInjMold -sim , an Open_Source solver using OpenFOAM “ is submitted at the Department of Mechanical Engineering at Indian Institute of Technology, Kharagpur in partial fulfillment of the requirements for the degree of Bachelor of Technology.

This thesis is based on independent work done by me under the supervision of Prof. Manab Kumar Das . Any contribution from other sources are acknowledged and referenced appropriately. The work has not been submitted to any other institute for any degree or diploma. I have conformed to the norms and guidelines given in the Ethical code of conduct.

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This is to certify that the project report entitled “ verification and validation of openInjMold -sim , an Open_Source solver using OpenFOAM” is submitted by Nikkula Krishnakarthik(22MF3IM29) to the Indian Institute of Technology Kharagpur for the Bachelor Thesis Project in Manufacturing Science and Engineering is a record of bona fide work carried out by him under my supervision and guidance during the academic year 2025-26.

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ABSTRACT

Injection molding (IM) is the most common process for manufacturing thermoplastic parts. Simulation of this process is an indispensable step in the design phase, but proprietary software licences are expensive and inflexible. In this study, the simulation of 3D, non-isothermal, non-Newtonian fluid flow of polymer melts is investigated using a solver in the open-source library OpenFOAM®.

The solver, **openInjMold**, is based on a compressible, two-phase flow model using a cell-centered unstructured finite volume method (FVM) and volume of fluid (VOF) technique. The rheological behaviour of the polymer melt is characterized by the **Cross-WLF model**, and equation of state is modeled using the modified **Tait equation**.

Keywords : injection molding, Cross-WLF model, Tait model, finite volume method, openInjMoldSim, OpenFOAM®

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Nomenclature

Symbol	Description
ρ	Fluid density
u	Velocity vector
p	Pressure field
T	Temperature field
σ	Cauchy stress tensor
τ	Deviatoric stress tensor
η	Fluid viscosity
$\dot{\gamma}$	Generalized shear rate
D	Deformation rate tensor
c_p	Specific heat capacity
k	Thermal conductivity
β	Compressibility coefficient
α	Phase fraction (VOF)
$\hat{V}$	Specific volume (PVT)
T_g	Glass transition temperature
η_0	Viscosity at null shear rate
τ^*	Cross-WLF shear stress constant
n	Power-law index
D_1, D_2, D_3	WLF model parameters
C_1, C_2	WLF model parameters
$b_1 \dots b_9$	Modified Tait model coefficients

Chapter 1

Introduction

1.1 Background and Motivation

The demand for plastic materials is ever-growing, increasing the requirements for transformation processes like injection molding (IM). IM is the most widely used process for manufacturing thermoplastic parts, as it allows for the production of components with a wide range of geometries and provides a good return on investment in the medium to long term. However, the IM process has high fixed costs associated with the molds and the large number of process variables that must be controlled and optimized. Due to these complexities, numerical simulation has become an indispensable step in the design phase. Simulation helps to minimize experimental iterations, decrease development costs, and ensure parts meet the desired geometrical and mechanical requirements.

1.2 Problem Statement

While several proprietary software solutions for IM simulation exist, they present two major drawbacks:

1. **High Cost:** The licenses for this software are expensive.
2. **Lack of Flexibility:** Proprietary software is typically a "black box" that cannot be adapted to specific user needs.

These limitations often prevent small to medium-sized companies from accessing these critical design tools. Open-source free alternatives, such as the computational fluid dynamics (CFD) library OpenFOAM® [**openfoam**], can overcome these limitations by providing access to adapt the source code directly.

1.3 Literature Review

OpenFOAM® has been successfully used in polymer processing areas like extrusion and blow molding. Its application to injection molding, however, has seen varied attempts.

- Araújo et al. [araujo2009] presented a 3D non-isothermal flow solver but treated both polymer and air as incompressible.
- Nagy and Steinbichler [nagy2019] introduced a framework for a *compressible* two-phase model, which is necessary to describe the filling, packing, and cooling stages.
- Magalhães [magalhaes2016] implemented the energy equation and Bird-Carreau rheological model into the interFoam solver.
- Recently, Mole et al. [mole2017] and Krebelj and Turk [krebelj2019] implemented both the **Cross-WLF** (rheology) and **modified Tait** (equation of state) models into the compressibleInterFoam solver. This new solver was designated **openInjMoldSim**.

Preliminary studies by Fontaínhas [fontainhas2019] used this solver for a simple 3D case and found results were qualitatively similar to Moldex3D® [moldex3d], but a rigorous mesh-convergence study was not completed.

1.4 Project Objectives

This work describes the rigorous numerical verification and validation of the open-source solver, openInjMoldSim. The primary objectives are:

1. To verify the numerical implementation of the solver against analytical solutions for a simple Newtonian flow case.
2. To perform detailed mesh sensitivity studies using Richardson’s extrapolation to ensure accurate and converged solutions.
3. To assess the effect of using a compressible vs. an incompressible formulation for the filling stage.
4. To compare the accuracy and computational performance of openInjMoldSim against the widely used proprietary software Moldex3D® for a complex industrial geometry.

1.5 Report Organization

This report is organized as follows:

- **Chapter 2** presents the mathematical formulation, including the governing equations, constitutive models (Cross-WLF), and equation of state (modified Tait) used in the solver.
- **Chapter 3** details the simulation methodology, material properties, boundary conditions, and validation metrics common to all case studies.
- **Chapter 4** presents the results and discussion from the three validation case studies.
- **Chapter 5** summarizes the main conclusions of this work and suggests directions for future research.

Chapter 2

Mathematical Modeling

The solver is based on a compressible, non-isothermal formulation, which provides a rigorous basis for simulating the subsequent packing and cooling stages of injection molding. The implementation is performed within the OpenFOAM® framework.

2.1 Governing Equations

The flow of the polymer melt and air during the filling stage is assumed to be immiscible and compressible. The flow is governed by the conservation equations for mass, momentum, and energy [pedro2020openinjmoldsim].

Mass Conservation (Continuity):

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho u) = 0 \quad (2.1)$$

Momentum Conservation:

$$\frac{\partial (\rho u)}{\partial t} + \nabla \cdot (\rho u u) = \nabla \cdot \sigma \quad (2.2)$$

Energy Conservation:

$$\frac{\partial (\rho c_p T)}{\partial t} + \nabla \cdot (\rho c_p u T) = \beta T \left(\frac{\partial p}{\partial t} + u \cdot \nabla p \right) + p \nabla \cdot u + \sigma : \nabla u + \nabla \cdot (k \nabla T) \quad (2.3)$$

Here, σ is the total Cauchy stress tensor, defined as $\sigma = -pI + \tau$, where τ is the deviatoric stress tensor. For a compressible generalized Newtonian fluid, τ is defined as

[pedro2020openinjmoldsim]:

$$\tau = 2\eta(T, p, \dot{\gamma})D - \frac{2}{3}\eta(T, p, \dot{\gamma})(\nabla \cdot u)I \quad (2.4)$$

2.2 Interface Capturing: Volume-of-Fluid (VOF)

To capture the location of the melt front, an advanced free-surface model based on the Volume-of-Fluid (VOF) method is used. This method solves a transport equation for the phase fraction, α , which ranges from 0 (air) to 1 (polymer melt) [pedro2020openinjmolidsim].

The physical properties (like density ρ or viscosity η) in these interface cells are calculated as a weighted average:

$$\psi = \psi_m \alpha + \psi_a (1 - \alpha) \quad (2.5)$$

where subscripts m and a denote melt and air, respectively.

The governing equation for α includes an artificial interface compression term to prevent numerical diffusion and maintain a sharp interface [pedro2020openinjmolidsim]:

$$\frac{\partial \alpha}{\partial t} + \nabla \cdot (\alpha u) + \nabla \cdot [\alpha(1 - \alpha)u_r] = S_p + S_u \quad (2.6)$$

Here, u_r is the "compression velocity," an artificial term that is non-zero only at the interface.

2.3 Constitutive Models (Rheology)

The rheological behavior defines the fluid's viscosity. The air is assumed to be a simple Newtonian fluid. The polymer melt, however, is a non-Newtonian fluid, and its behavior is described by the **Cross-WLF model**.

2.3.1 The Cross-WLF Model

This model is widely used for IM simulations and describes the polymer's viscosity η_m as a function of temperature (T), pressure (p), and shear rate ($\dot{\gamma}$) [chiang1991].

It is a combination of two models:

1. **The Cross Model:** This describes the shear-thinning (pseudoplastic) behavior, where viscosity decreases as the shear rate increases [pedro2020openinjmolidsim].

$$\eta_m(T, p, \dot{\gamma}) = \frac{\eta_0(T, p)}{1 + (\eta_0(T, p)\dot{\gamma}/\tau^*)^{1-n}} \quad (2.7)$$

Here, n is the power-law index, τ^* is the shear stress constant, and η_0 is the viscosity at a null shear rate.

2. **The WLF Model:** This model is used to calculate η_0 , describing its strong dependence on temperature and pressure [wlf1955].

$$\eta_0(T, p) = D_1 \exp \left[\frac{-C_1(T - T_0)}{C_2 + T - T_0} \right] \quad (2.8)$$

where $T_0 = D_2 + D_3 p$. The terms D_1, D_2, D_3, C_1 , and C_2 are material-specific parameters.

2.4 Equation of State (PVT Behavior)

The pressure-volume-temperature (PVT) relationship describes how the material's density changes. Polymers shrink as pressure increases, and they expand as temperature increases.

2.4.1 The Modified Tait Model

The most common model to describe the PVT behavior of polymers is the modified Tait model [pedro2020openinjmoldsim]. The specific volume, $\hat{V}$ (which is $1/\rho$), is given by:

$$\hat{V} = \hat{V}_0 \left[1 - C \ln \left(1 + \left(\frac{P}{B} \right) \right) \right] + \hat{V}_t \quad (2.9)$$

where $C = 0.0894$ is a constant. The terms $\hat{V}_0$, B , and $\hat{V}_t$ are complex functions of temperature and pressure, defined differently above and below the material's transition temperature (T_g or T_m). These functions depend on the material parameters b_1 through b_9 .

2.5 Numerical Discretization

The governing equations are solved numerically using a collocated **Finite Volume Method (FVM)**. The implementation is done using the OpenFOAM® computational library [pedro2020openinjmoldsim].

Chapter 3

Methodology and Validation Setup

This project uses three distinct case studies to verify and validate the openInjMoldSim solver. This chapter describes the common material properties, boundary conditions, and analysis methods used.

3.1 Material Properties

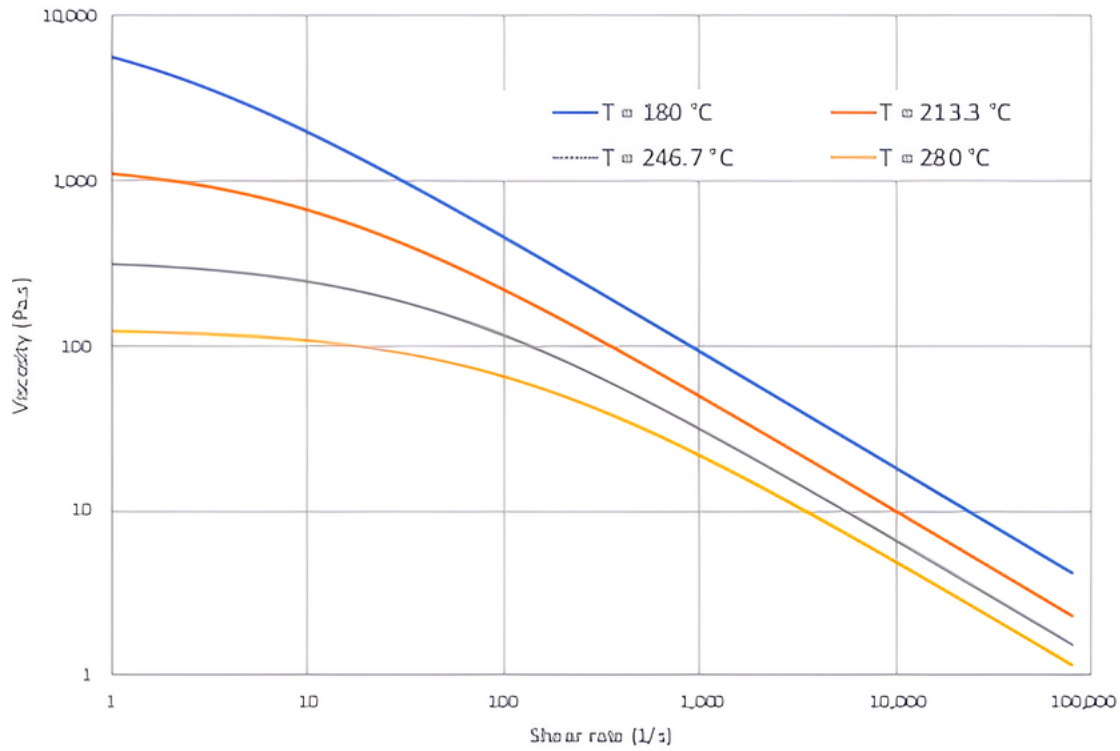
- **Polymer Melt:** The material used is GPPS Styron 678.
 - **Rheology:** Cross-WLF coefficients are in Table 3.1. Viscosity behavior is in Figure 3.1.
 - **PVT:** Modified Tait model coefficients are in Table 3.2. PVT behavior is in Figure 3.2.
 - **Thermal:** $c_p = 2100 \text{ J kg}^{-1} \text{ K}^{-1}$ and $k = 0.15 \text{ W m}^{-1} \text{ K}^{-1}$.
- **Air:** Treated as a Newtonian fluid with $\eta = 0.1 \text{ Pas}$.
 - **Thermal:** $c_p = 1007 \text{ J kg}^{-1} \text{ K}^{-1}$ and $k = 0.0263 \text{ W m}^{-1} \text{ K}^{-1}$.

Table 3.1: Cross-WLF coefficients for GPPS Styron 678 [pedro2020openinjmoldsim].

Parameter	Value
n	0.2903
τ^* (Pa)	13,678
D_1 (Pa.s)	7.44×10^{10}
D_2 (K)	373.15
D_3 (K/Pa)	0.0
A_1 (C_1)	25.971
A_2 (C_2) (K)	51.6

Table 3.2: Modified Tait model coefficients for GPPS Styron 678 [pedro2020openinjmoldsim].

Parameter	Value
b_{1m} (m ³ /kg)	0.0009881
b_{2m} (m ³ /(kg.K))	$7.03 \cdot 10^{-7}$
b_{3m} (Pa)	$1.71 \cdot 10^8$
b_{4m} (1/K)	0.004495
b_{1s} (m ³ /kg)	0.0009873
b_{2s} (m ³ /(kg.K))	$2.89 \cdot 10^{-7}$
b_{3s} (Pa)	$2.43 \cdot 10^8$
b_{4s} (1/K)	0.003106
b_5 (K)	373.98
b_6 (K/Pa)	$2.88 \cdot 10^{-7}$
b_7 (m ³ /kg)	0
b_8 (1/K)	0
b_9 (1/Pa)	0



3.2 Common Simulation Setup

For all case studies, the initial and boundary conditions are summarized in Table 3.3.

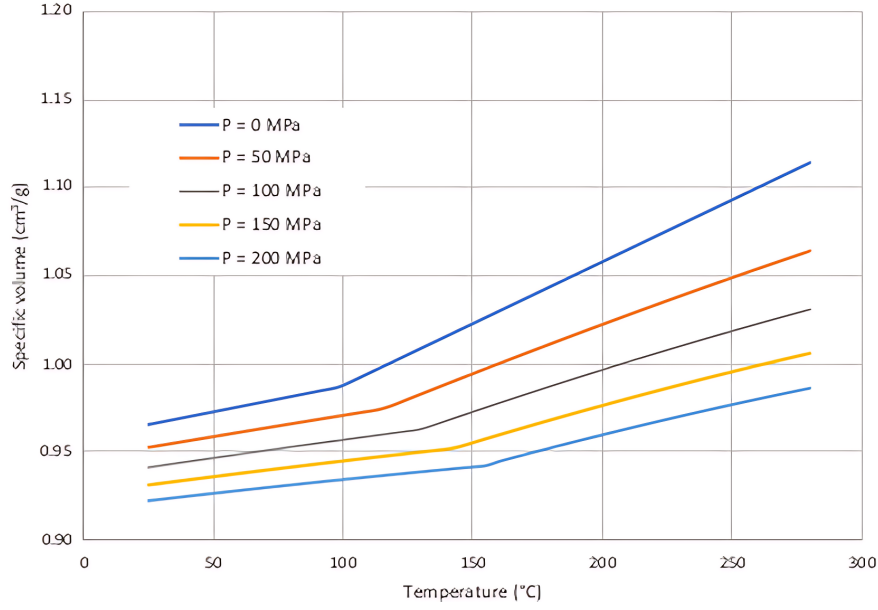


Figure 3.2: Variation of the polymer specific volume with temperature and pressure fields for the GPPS Styron 678 (based on Fig. 2 in [pedro2020openinjmoldsim]).

Table 3.3: Initial and boundary conditions for all case studies [pedro2020openinjmoldsim].

	Temperature (T)	Velocity (U)	Pressure (P)	Phase (a)
Inlet	230 °C	u_{in}	Null normal grad.	1
Walls	Convection $T_{mold} = 50^\circ\text{C}$ $h = 1250\text{W}/(\text{m}^2\text{K})$	Null velocity	Null normal grad.	Null normal grad.
Outlet	Null normal grad.	Null normal grad.	$1 \times 10^5\text{ Pa (atmos.)}$	Null normal grad.
Initial	230 °C	Null velocity	$1 \times 10^5\text{ Pa (atmos.)}$	0

3.3 Verification and Validation Metrics

3.3.1 Switch-Over Time

The switch-over time (t_{SO}) is the time taken to fill **98%** of the total cavity volume. For an incompressible fluid, this time can be calculated analytically [pedro2020openinjmoldsim]:

$$t_{SO} = \frac{V}{A \times U_{in}} \quad (3.1)$$

where V is 98% of the cavity volume, A is the inlet area, and U_{in} is the inlet velocity.

3.3.2 Interface Compression (IC/NIC)

Simulations were run with the VOF artificial interface compression term considered (**IC**) and neglected (**NIC**) to assess its impact on mass conservation [pedro2020openinjmoldsim].

3.3.3 Discretization Error (Richardson's Extrapolation)

This method is used to estimate the discretization error of a numerical solution. It uses results from simulations on three different meshes (coarse, medium, fine) to extrapolate the value of a variable to an infinitely refined mesh. See [Appendix A](#) for details.

Chapter 4

Results and Discussion

4.1 Case Study 1: Filling of a Cylindrical Cavity

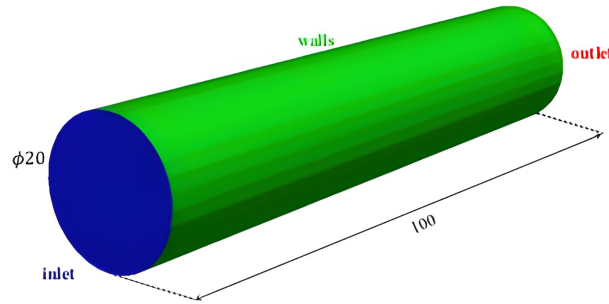


Figure 4.1: Geometry of the cylindrical cavity for Case Study 1 (based on Fig. 3 in [pedro2020openinjmolddsim]).

4.1.1 Objective and Setup

This case study involves the flow of a **Newtonian fluid** (Cross-WLF with $n = 1$) along a cylindrical channel. The goal is to verify the solver's basic implementation against a known analytical solution. Three mesh densities were used (M1, M2, M3).

4.1.2 Findings: Interface Compression (IC/NIC)

The switch-over time (t_{SO}) was calculated and compared to the analytical value of 0.0245 s.

- **With Interface Compression (IC):** The IC variation gave very accurate results for the coarse (M1) and medium (M2) meshes, with errors of 1.6% and 0.4%, respectively [pedro2020openinjmoldsim].
- **Without Interface Compression (NIC):** The NIC variation gave a massive 88.6% error on the coarse mesh (M1) [pedro2020openinjmoldsim].
- **Conclusion:** The artificial interface compression term (IC) is crucial for coarse and medium meshes as it minimizes interface diffusion and ensures mass conservation [pedro2020openinjmoldsim].

4.1.3 Findings: Compressible vs. Incompressible

The simulation was run using both the full compressible modified Tait model and a simplified incompressible model.

- **Switch-Over Time:** The times were nearly identical (0.4% difference on M3 mesh) [pedro2020openinjmoldsim].
- **Velocity & Pressure:** The predicted velocity and pressure fields were very close [pedro2020openinjmoldsim].
- **Conclusion:** For a simple Newtonian flow, using an incompressible formulation does not significantly affect the accuracy of the final results for the filling stage [pedro2020openinjmoldsim].

4.2 Case Study 2: Rectangular Cavity with Insert

4.2.1 Objective and Setup

This case study uses the non-Newtonian (Cross-WLF) material to fill a rectangular cavity with a cylindrical insert. This geometry is more complex, causing the flow front to split and then merge, forming a **weld line** [pedro2020openinjmoldsim]. The main objective was to compare the compressible and incompressible formulations for a more realistic non-Newtonian flow.

4.2.2 Findings: Compressible vs. Incompressible

The velocity, temperature, and pressure fields were analyzed at the switch-over point for the most refined mesh (M3).

- **Flow Fields:** There were **no visible differences** in the velocity or temperature distributions between the two formulations [pedro2020openinjmoldsim].

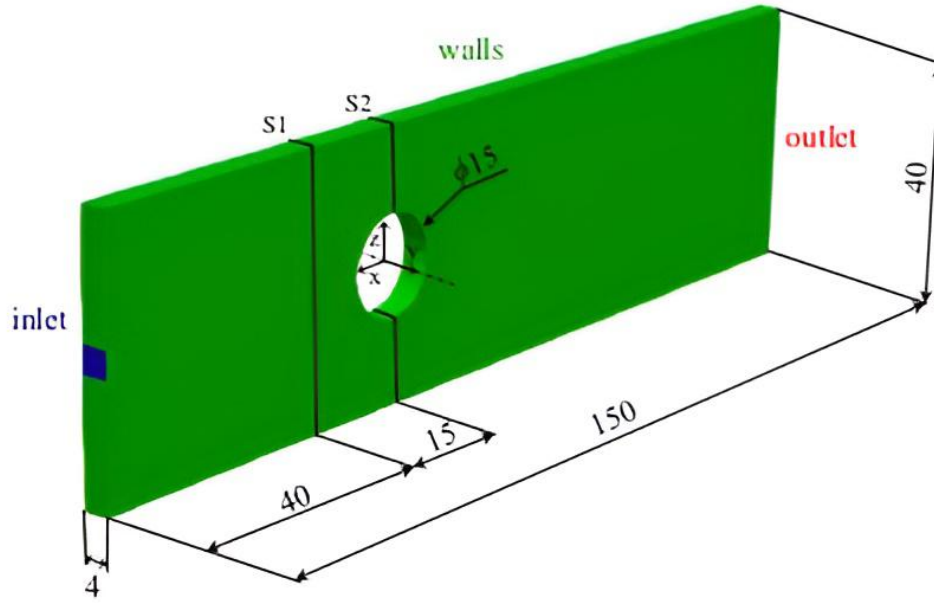


Figure 4.2: Geometry and boundary patches for Case Study 2 (based on Fig. 4 in [pedro2020openinjmoldsim]).

- **Quantitative Analysis:** The relative differences in average and maximum temperature and velocity were negligible, generally less than 4% [pedro2020openinjmoldsim]. The maximum pressure required to fill the cavity was also nearly identical (0.9
- **Performance:** The incompressible formulation was computationally cheaper (5% faster) [pedro2020openinjmoldsim].
- **Conclusion:** The incompressible formulation is a good and efficient approximation for simulating the *filling stage* of the IM process [pedro2020openinjmoldsim].

4.3 Case Study 3: Filling of a Tensile Test Specimen

4.3.1 Objective and Setup

This is the most complex case, simulating the filling of a standard tensile test specimen, including its feeding system (sprue, runner, and gate) [pedro2020openinjmoldsim]. The objective was to compare the accuracy and performance of openInjMoldSim with the proprietary software Moldex3D® [pedro2020openinjmoldsim].

4.3.2 Mesh Sensitivity Analysis

A mesh sensitivity study was performed on both solvers using the Richardson's extrapolation (RE) method [pedro2020openinjmoldsim].

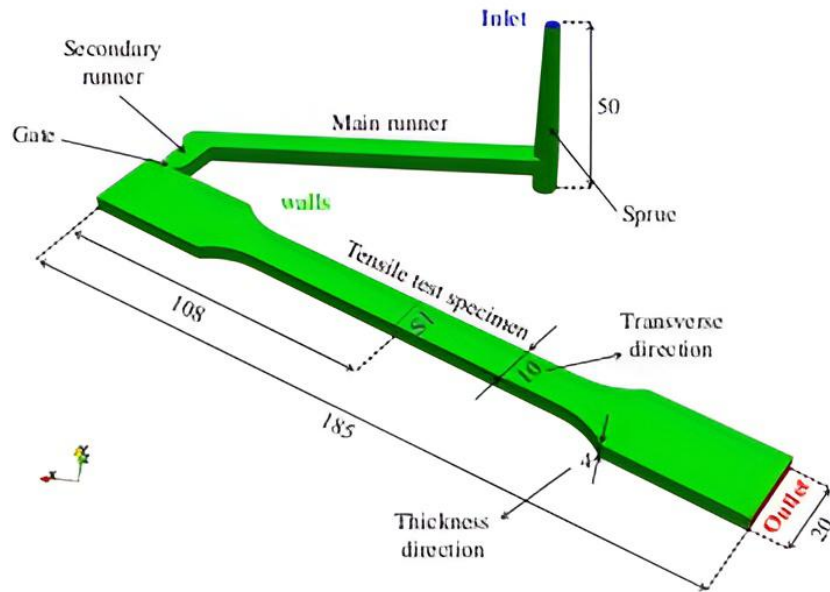


Figure 4.3: Geometry and boundary walls for Case Study 3 (Tensile Test Specimen) (based on Fig. 8 in [pedro2020openinjmoldsim]).

- **openInjMoldSim:** Converged results (relative error < 10%) were achieved with mesh M3 (2.1 million cells) [pedro2020openinjmoldsim].
- **Moldex3D®:** The error for mesh M3 was still very high (23.8%). An additional, much finer mesh, M4 (15.3 million cells), was required to reach the same order of accuracy [pedro2020openinjmoldsim].
- **Conclusion:** openInjMoldSim required significantly fewer cells to achieve a converged, accurate solution compared to Moldex3D® [pedro2020openinjmoldsim].

4.3.3 Comparison of Flow Front and Pressure

- **Flow Front:** The melt front predicted by openInjMoldSim showed a realistic, rounded "fountain flow" profile. In contrast, the Moldex3D® front appeared as an unrealistic "plug-like" surface [pedro2020openinjmoldsim].
- **Pressure Evolution:** openInjMoldSim correctly predicted a sharp change in the pressure slope when the melt passed through the thin **gate**. Moldex3D® **failed to capture** this critical physical event and showed nonphysical oscillations [pedro2020openinjmoldsim].
- **Conclusion:** The openInjMoldSim solver predicted more realistic physical phenomena than the proprietary software [pedro2020openinjmoldsim].

4.3.4 Accuracy and Performance Comparison

- **Accuracy:** For any given mesh level, openInjMoldSim consistently produced more accurate results (lower error) for pressure and velocity than Moldex3D® [pedro2020openinjmoldsim].
- **Performance (Per-Accuracy):** To get the same accuracy, one must compare **M3 for openInjMoldSim** with **M4 for Moldex3D®**. The proprietary software was only 1.4 times faster, an advantage that can be overcome by openInjMoldSim's unrestricted parallelization [pedro2020openinjmoldsim].
- **Conclusion:** The performance is comparable when aiming for the same accuracy. openInjMoldSim's lack of licensing restrictions allows it to match proprietary speed if sufficient hardware is available [pedro2020openinjmoldsim].

Chapter 5

Conclusions and Future Work

5.1 Conclusions

This project successfully verified and validated the open-source solver openInjMoldSim [pedro2020openinjmoldsim]. The main conclusions are:

1. **VOF Formulation:** The artificial interface compression (IC) term is essential for obtaining accurate results on coarse and medium meshes [pedro2020openinjmoldsim].
2. **Compressible vs. Incompressible:** For the filling stage, using a full compressible formulation provides no significant accuracy benefit over a simpler incompressible formulation, which is more efficient [pedro2020openinjmoldsim].
3. **Accuracy vs. Moldex3D®:** openInjMoldSim demonstrates **better accuracy** than Moldex3D®. It achieves mesh-converged results with one fewer level of refinement and captures critical physical phenomena that Moldex3D® missed [pedro2020openinjmoldsim].
4. **Performance vs. Moldex3D®:** When compared on an **equal-accuracy basis**, openInjMoldSim's performance is comparable, especially given its ability to scale with unlimited cores [pedro2020openinjmoldsim].

In summary, this work shows that the open-source solver openInjMoldSim is a highly accurate and viable alternative to expensive proprietary software for modeling the IM filling stage [pedro2020openinjmoldsim].

5.2 Future Work

Based on the findings, the following future work is recommended:

1. **Performance Improvement:** Additional studies should be made to improve the solver's calculation performance and efficiency [pedro2020openinjmoldsim].

2. **IC Term Criteria:** A detailed study should be performed to identify the exact criteria for when the artificial interface compression term should be used [pedro2020openinjmoldsim].
3. **Packing and Cooling:** The solver's capabilities should be extended and validated for the subsequent **packing and cooling stages**.

Appendix A

Richardson's Extrapolation

Richardson's extrapolation (RE) is a method to estimate the discretization error in CFD [celik2008]. When the values of a property (θ) are known for three meshes (e.g., θ_1 on coarse, θ_2 on medium, θ_3 on fine) with a constant grid refinement factor (R), the extrapolated value (θ_{ext}) for an infinitely fine mesh can be estimated.

The apparent order, ap , is first calculated as [pedro2020openinjmoldsim]:

$$ap = \frac{\ln |(\theta_3 - \theta_2)/(\theta_2 - \theta_1)|}{\ln R} \quad (\text{A.1})$$

The extrapolated value is then given by [pedro2020openinjmoldsim]:

$$\theta_{ext} = \frac{R^{ap} \theta_2 - \theta_3}{R^{ap} - 1} \quad (\text{A.2})$$

Finally, the relative error for any mesh i is estimated by [pedro2020openinjmoldsim]:

$$e_i(\%) = \frac{\theta_i - \theta_{ext}}{\theta_{ext}} \times 100 \quad (\text{A.3})$$

In this work, $R = 2$ was used. Results were considered converged when the relative error was below 5% [pedro2020openinjmoldsim].